

ARTICLE



A computational framework for agent-based assessment of multiple environmental exposures

Oliver Schmitz¹✉, Kees de Hoogh^{2,3}, Nicole Probst-Hensch^{2,3}, Ayoung Jeong^{2,3}, Benjamin Flückiger^{2,3}, Meng Lu⁴, Aisha Ndiaye⁵, Danielle Vienneau^{2,3}, Gerard Hoek⁵, Kalliopi Kyriakou⁵, Roel C. H. Vermeulen^{5,6} and Derek Karssen¹

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BACKGROUND: Agent-based assessment of long-term personal exposure to environmental factors accounts for spatio-temporal variation in exposures along daily activity tracks of individuals. Application up to nationwide study populations requires integration of large data sets on environmental factors, personal behavior, and socio-economic status, as well as propagating uncertainties in these inputs to personal exposure values.

OBJECTIVE: To develop and illustrate a methodology and software framework for agent-based personal exposure assessment for large cohorts, including uncertainty assessment.

METHODS: We design an agent-based methodology that addresses the sparse information on individual activity patterns available in large cohorts. This methodology was implemented in a Python-based open-source and reusable framework, which was subsequently applied to assess exposure to air pollution and noise for 626,381 residential addresses in the province of Utrecht, the Netherlands. Air pollution exposures were also assessed across all addresses in Switzerland and the EPIC-NL cohort in the Netherlands.

RESULTS: The designed framework aggregates time by divisions marked by a particular pattern in individual movement (e.g., weekdays, weekend days). Movement over a division is represented by a sequence of activities, each with a duration and spatial context, i.e., the geographical area where the activity takes place. Several activity types are included, each with a methodology to assess the spatial context, for instance, the route from home to work location. Uncertainty in inputs is defined by probability distributions constrained by observational data, if available, like statistics on origin and destination of trips, and propagated to calculated personal exposures through Monte Carlo simulation. The exposures assessed through our framework result in minor to moderate differences with those calculated using home-based exposure (for Utrecht an r^2 of 0.79 for noise and 0.98 for nitrogen dioxide (NO₂) and particulate matter with aerodynamic diameters of 2.5 microns or smaller (PM_{2.5}), respectively), in particular leading to reduced contrast across the population in exposures.

IMPACT: Epidemiological studies on long-term effects of air pollution typically use a residential-based exposure assessment. However, it fails to account for individual mobility and spatial contrasts in environmental concentrations. While there is thus a need to investigate activity-based methods, their implementation is constrained by the lack of conceptual frameworks and software, particularly for large cohorts, which present unique demands regarding data inputs and computation. To address this gap, we introduce general concepts and a reusable, open-source software framework, designed for cluster computing, that can be applied consistently across a wide array of environmental factors and cohort studies.

Keywords: Activity-based modeling; Personal exposures; Air pollution; Noise; Uncertainty estimation

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INTRODUCTION

High-quality and individual-level assessment or modeling of environmental factors is essential for, and underpins, studies evaluating the effects of environmental factors on human health. For example, long-term exposure to air pollution may contribute to the development of asthma, lung cancer, diabetes, and cardiovascular disease [e.g., [1–4]], noise is associated with an increased mortality risk [e.g., [5, 6]], while exposure to green or

blue space is generally considered beneficial for human health [e.g., [7]]. Identified relationships can be used to quantify contributions of exposure to population health or disease, while they can lead to improved health management [e.g., [8, 9]] or urban planning considering healthier living environments [e.g., [10–14]].

The range, magnitude, and detail of available environmental data have increased considerably, which contributes to improved

¹Department of Physical Geography, Faculty of Geosciences, Utrecht University, Utrecht, The Netherlands. ²Swiss Tropical and Public Health Institute, Allschwil, Switzerland.

³University of Basel, Basel, Switzerland. ⁴Department of Geography, University of Bayreuth, Bayreuth, Germany. ⁵Utrecht University, Institute for Risk Assessment Sciences, Division of Environmental Epidemiology, Utrecht, The Netherlands. ⁶University Medical Centre Utrecht, Utrecht, Netherlands. ✉email: o.schmitz@uu.nl

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assessment of human exposures. However, several important challenges remain. Here, we provide a short background and briefly review those challenges that are directly addressed by our proposed framework for modeling personal exposure assessment. Individual-level exposures to environmental factors can be assessed by direct measurement of personal exposures or by using some spatio-temporal model that combines locations of individuals with existing environmental data [e.g., [15–18]]. Direct measurement of exposures requires equipment that continuously samples the environmental factor and is carried by each individual. While applicable for detailed analysis, it becomes quickly unfeasible with large cohorts due the practical issues and costs involved [e.g., [17]]. For large-scale epidemiological studies, which are also the focus of this study, exposure is thus mostly assessed through a model that combines the location of an individual with existing environmental factor data compiled for the complete study area, for example, by using long-term stationary air quality data [e.g., [19, 20]]. Typically, only the residential location is known and used in such studies.

For our model-based approach, the first major challenge is accounting for human mobility in exposure assessment [21]. Assessing exposures at residential locations has resulted in an improved understanding of relationships between environmental stressors and health effects [e.g., [22–24]], but estimates at the residential location may not be representative for exposures integrated over a longer time (e.g., a year) as individuals are mobile and thus exposed to different exposure levels at different locations and moments in time. Hence, several studies have stressed the need and evaluated methods for incorporating mobility of individuals in exposure assessment [e.g., [25–28]]. As GPS tracking data for a large number of persons is often not available and impractical, these approaches mostly rely on agent-based modeling [29–31] or related methods modeling the movements of individuals over time. In the following, we refer to a person when using the more general term agent.

Personal exposures are then calculated by intersecting the space-time track of an individual with spatio-temporal environmental data, integrating exposures over time. This mobility-based exposure assessment has most widely been applied to small cohorts [e.g., [27, 28, 32–36]], mainly because the methodology requires rich data to characterize mobility patterns at the individual level. This information is often not available in a deterministic fashion for large cohorts, let alone for the complete population of a larger region or country. In addition to the sparse data availability for large populations, mobility-based exposure assessment may become infeasible because of the massive run times associated with large-scale agent-based modeling.

Another challenge—in the framework of the exposome [4, 37–39], i.e., the totality of human environmental exposures from conception onwards—is the need for assessment of exposures to a range of environmental factors, preferably harmonized, for evaluating associations between environment and health outcomes.

The third and final important challenge addressed by our framework is the quantification of uncertainty in personal exposures. Both environmental factor surfaces and space-time tracks of individuals come with uncertainties. For epidemiological analysis, it is beneficial to propagate these input uncertainties towards the output of the exposure assessment [e.g., [40–44]], i.e., the calculated personal exposure value.

We address these challenges by proposing a general methodological framework for mobility-based exposure assessment, and further designed and implemented a software solution. Our purposely designed software framework will contribute to standardization and reproducibility of research, while it may ease the implementation of exposure assessment in future studies. The framework provides functionality for the specification of agents' activity diaries, including the duration of activities and their spatial

contexts, that is, the location of a person during an activity. The software framework currently supports three different spatial contexts: for known locations (e.g., residential address location), unknown locations within a certain area (e.g., sporting in the neighborhood), and locations along a route (e.g., while commuting). The framework enables integrated assessment of multiple environmental factors, calculating for each individual the exposure over time and the mean exposure over a certain time span. The framework is designed for exposure assessment for large cohorts or nationwide administrative cohorts, using probability distributions to determine, for example, the travel mode and duration of an activity from the literature or statistics (e.g., from time-activity diary data) for the studied population, including subsets (e.g., considering socio-economic profiles) to inform the spatio-temporal behavior. We demonstrate this approach in a case study for a nationwide personal exposure assessment. To enable this probabilistic approach, the framework allows for defining most of the inputs and parameters as probability distributions, solving the error propagation problem through Monte Carlo simulation. This also results in the assessment of uncertainty associated with the calculated personal exposures for each individual separately.

Here, we describe, illustrate, and evaluate this conceptual framework and its software implementation. “Conceptual model” describes the concepts used; in “Software implementation” we introduce a software implementation of the modeling framework, which demonstrates the construction of activity profiles for agents, simulation setup, and further steps for model assessment. In “Case study” we illustrate the application of the framework in an exposure assessment case study, where we estimate exposures to air pollution and noise for each residential address in the province of Utrecht, the Netherlands. We compare estimates for two different activity profiles, i.e., for homemaker and commuter. Using the same approach, we derived personal exposure estimates for NO₂ and PM_{2.5} for Switzerland (Swiss National Cohort [SNC, [45, 46]] and the Study on Air Pollution and Lung and Heart Diseases in Adults [SAPALDIA, [47–49]]) and the Dutch contribution to the European Prospective Investigation into Cancer study ([EPIC-NL, [50]]) for the separate research project ‘Accounting for mobility in air pollution exposure estimates in studies on long-term health effects (MOBI-AIR)’ where considering mobility in individual exposure estimates is leading to less biased results in health studies is investigated. This methodological framework was developed as part of the MOBI-AIR project, which investigated whether there is a bias in long-term air pollution health studies if residential-based exposures are used. As part of MOBI-AIR, the approach presented here was used to calculate mobility-enhanced air pollution exposures for NO₂ and PM_{2.5} for cohorts in the Netherlands (EPIC-NL [50]) and Switzerland (SNC [45, 46] and SAPALDIA [47–49]), where differences in health effects between residential and mobility-enhanced exposures were analyzed [e.g., [51]]. We conclude with a discussion of the capabilities of the presented open-source software framework.

CONCEPTUAL MODEL

General framework

Long-term personal exposure can be defined as the contact of a human or agent with the environment over a long time (e.g., years or longer). In theory, the assessment of long-term exposure would require determining all movements of a person over time as well as acquiring multiple environmental factor values for all locations visited by the person. Combined with activity factors for common activities, this would allow the calculation of integrated personal exposures [52]. In practice, this is not feasible as people perform a large number of activities, such as commuting, exercising, gardening, and each activity can be associated with changes in the agent's location. This level of detail is not available for exposure assessments aimed at large cohorts, over time spans of

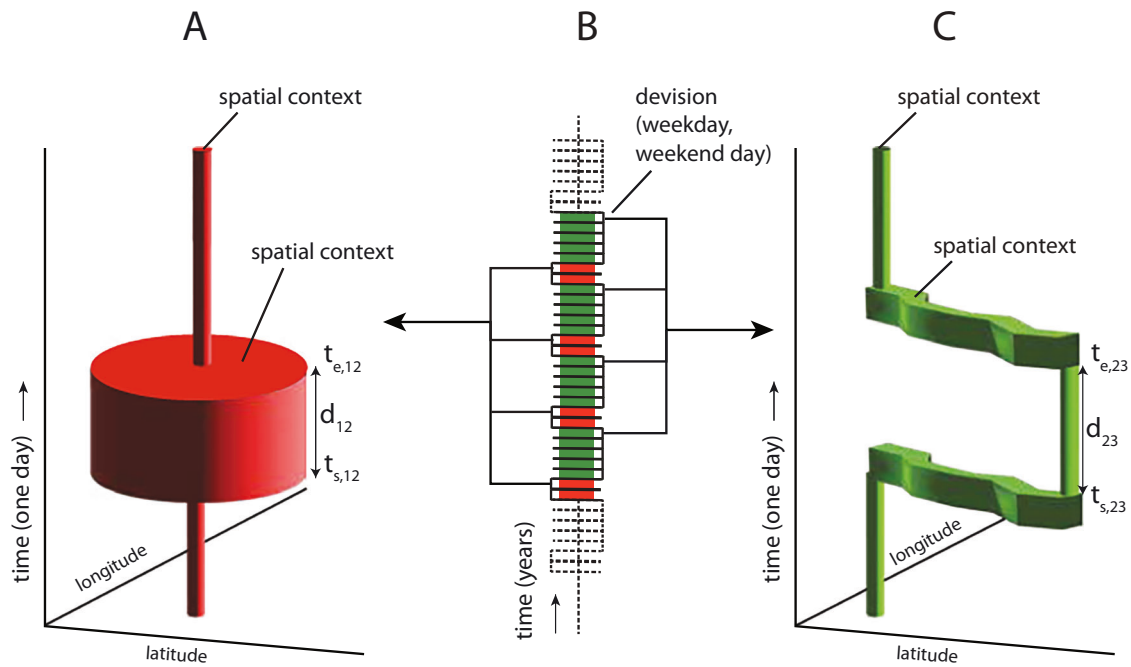


Fig. 1 Schematic representation of the conceptual model. **A** Space-time activity of one agent over the division with five activities defined in the agent's diary. One division is shown here for a weekday. Included are two commute activities along the traffic infrastructure, back and forth to the agent's work location. The work activity lasts duration d_{13} specified by $t_{s,13}$ and $t_{e,13}$ as start and end times, respectively. **B** Long term time span for which exposure assessment is representative (only part of time axis shown), subdivided in two divisions for weekdays and weekend days. **C** Space-time activity of one agent over a weekend division, with three activities defined in the agent's diary. The second activity uses a larger radius for the circular spatial context, e.g., to represent an outdoor activity in the surroundings.

several years. In addition, a detailed simulation of each agent's space-time path would be far too time-consuming to compute.

In our conceptual framework, we thus model personal exposures by aggregating time and activities [e.g., [15, 53]] as space-time volumes [54], as shown in Fig. 1, assuming that the person visits each geographical location within the volume for the same duration. Time is aggregated by defining I time divisions $i = 1..I$ each marked by a particular pattern in individual movement and environmental factor values (Fig. 1A). In the example case presented in the second part of this article, we divide environmental factor values into weekdays and weekend days, as, for example, air pollution patterns differ between these two divisions [55, 56]. In addition, activity profiles of an agent were used to create divisions, for example, representing a weekday with commuting trips or one with home-based work. It is assumed that movement of individuals and diurnal variation in environmental factor values remain the same across all instances of each specific division, e.g., in a year, all weekdays with commuting will result in the same exposure for the person considered. The agent-based model will only run over the duration of the division used, here 24 h, resulting in an exposure value that is considered representative for that division across the time [54].

Secondly, movement of the individual over the time span of a division is aggregated by representing it as a sequence of activities $j_i = 1..J_i$, with J_i the number of activities in the division i (e.g., a weekday with commuting) (Fig. 1B). Each activity has an activity type, for instance "commute trip" and a duration $d_{ij} = t_{e,ij} - t_{s,ij}$, with $t_{s,ij}$, $t_{e,ij}$, the start and end time of the activity ij . The duration of an activity is typically between a few minutes and several hours. In addition, an activity has a spatial context defining the geographical area where the activity takes place. It is assumed the person is situated at all locations within the spatial context for the same duration between the start and end time of the activity. The personal exposure E_i over the division i (e.g., a weekday with

commute trip) is calculated as:

$$E_i = \frac{\sum_{j=1}^{J_i} E_{ij} \cdot d_{ij}}{\sum_{j=1}^{J_i} d_{ij}} \quad (1)$$

The long-term exposure E of the person is the time-weighted average of the exposure for the divisions:

$$E = \sum_{i=1}^I E_i \cdot p_i \quad (2)$$

In Eq. 2, p_i is the proportion of the time spent in each division. For example, a person who is commuting for 3 days in the week, works from home for 2 days, and has 2 days off in the weekend, three divisions would be used (commute weekdays, homebased workdays, weekend with leisure time), with a value of p_i of 3/7, 2/7, and 2/7, respectively, where 7 represents the number of days in the week. This approach would neglect days with other activities (e.g., holidays).

Activity diary and diary generation

An activity diary is used to describe the sequence of activities of an agent over the duration of a division. For each activity, the activity diary defines the start time $t_{s,ij}$ of the activity, its duration, the method to calculate the spatial context of the activity, and a number of input values for the spatial context calculation. Currently, three spatial context calculations are used:

- **Location.** A single geographical location, such as the residential or work location. Requires the geographical coordinates of the location as input.
- **Circular area,** an area centered around a geographical location. This spatial context can be used to model, e.g., shopping or recreational activities around the residential location. Requires as input the geographical coordinates of the center as well as the radius of the circular area.

- *Route*, the shortest route between two locations along traffic infrastructure. This spatial context can be used, e.g., to model commuting between residential and work locations. Requires as input geographical coordinates for start and end location of route, a mode of transport, and a traffic infrastructure network for each mode of transport.

In a fully deterministic model, where all inputs are known, the activity diary of each person can be directly read from a database. In most cases, however, inputs are not exactly known, and an activity diary generator is used to create different simulation runs, hereinafter referred to as realizations, of activity diaries (Fig. 1), and exposure is assessed for each realization. The configuration of the activity diary depends on which inputs are represented by stochastic variables and, in the current framework, includes the following capabilities:

- Drawing realizations of scalar values, for instance, for the radius of a *circular area* spatial context.
- Drawing spatial locations of the work location by random selection of a location in the study area or constrained by origin-destination matrices derived from travel survey data. Here, origin-destination matrices are used, providing for each origin area (e.g., an administrative unit or postal code area) the frequency of commute trips to destination areas. The framework randomly draws a destination area from this matrix and within this unit, a random location labeled as a possible work location (e.g., office building).
- Calculation of the mode of transport for an *activity* with a *route* spatial context from the Euclidian distance between start and end locations of route, as well as its duration for this mode of transport, over the traffic infrastructure network.

Numerical solution

All agents perform their activities within a common spatial and temporal extent, the simulation area, and the time duration of a *division*. The common spatial extent and discretisation are defined by the environmental attributes and cover, e.g., a province or a country at a particular raster cell size, e.g., 100 m. The common temporal extent is discretised with a temporal resolution suitable for combining activities and environmental factors, for example, 24 h are discretised into time steps of one minute.

Activities and environmental factors are combined to calculate personal exposures of each agent. Therefore, a common timeline between the agent's activities and the environmental factors is generated, i.e., at each change of an activity or factor, a new time step is introduced. For each of these time steps, the exposure is assessed by averaging the environmental attribute values within the spatial context of an activity. Lastly, the time-weighted average of all time steps is computed as overall exposure.

The variables in Eqs. 1 and 2 can be deterministic or stochastic variables. The latter is the case, for example, when durations d_{ij} are not exactly known and taken as probability distributions. In this case, uncertainties are propagated to the output by using Monte Carlo simulation, with realizations of the uncertain inputs or parameters created by the diary generation mechanism as input to the Monte Carlo simulation. This allows for calculating the complete probability distribution of E , for each agent.

SOFTWARE IMPLEMENTATION

The introduced concepts were implemented in an exposure assessment framework using the Python [57] programming language. The framework classes and methods provide building blocks for environmental scientists as model builders to develop models for reproducible exposure assessment. Two Python classes are intended to be used by model developers to describe the

activity diaries and diurnal cycles for all agents. The remaining classes are used internally by the framework, one class to generate spatial contexts, and one class to combine spatial contexts and environmental attributes to calculate personal exposures. Specifics of the modeling framework are described in the "Software availability" section.

Framework classes for model building

The diary of an agent consists of one or more activities that need to be specified by model builders. The modeling framework starts a diary from the simulation start time and derives consecutively start and end times for each activity. Error checking is included to ensure that all activities fit into a *division*.

The currently implemented activity classes derive from an abstract base class *Activity* holding general attributes for all activities, such as a unique identifier, the duration of an activity, and a description that can be defined by the model builder. The modeling framework provides three classes representing the spatial context types described in the Concepts section. The *Location* is set with a coordinate pair for the point location. The *Circular* uses a coordinate pair for the center location of the circular area and a radius. Within the buffer area, the mean value of an environmental factor will be calculated. The *Route* uses two coordinate pairs for the start and end locations and the mode of transport. The model builder can select between car, bike, by foot, or by train. The mean exposure value along a route will be calculated.

Figure 2 shows the application of framework methods to define a set of activities for an agent, and outlines the implementation of a weekday for a commuter and its execution. Lines 7–9 define the durations of an activity, which can be either known, set with an end time sampled from an interval, e.g., between 6:30 and 7:30, or derived as an estimate from the commute distance and a specific commute mode. Lines 10–12 define an activity of each spatial context type with the required arguments. Lines 13–16 specify the diary and included set of activities.

Agent diaries are embodied in a class derived from the *Diurnal* class. This class provides a set of methods that assist model builders in the calculation of activities, such as access to the routing engine or to query an origin-destination matrix. This and the collection of all agent diaries need to be initialized (lines 4–5). Afterwards, for each agent, the diary is specified and added to the collection (lines 6–16). For executing a simulation run, a random number generator, an index, and the time interval are assigned. The random seed can be fixed to obtain reproducible results. Furthermore, the activity diaries are generated, and exposure estimates are accordingly generated for a set of environmental factors (lines 20–22). A Monte Carlo ensemble can be created by a repeated execution of diary generation and exposure assessment.

Framework classes for exposure assessment

We implemented a *SpatialContext* class that is instantiated with a LUE dataset [58]. The LUE data format enables storage of environmental attributes with multiple spatial and temporal discretisations in a single dataset. We further implemented methods that are used to determine the spatial contexts, i.e., the bounding boxes and Boolean true values for the occurrences of an activity. These contexts correspond to a raster cell for a given *Location* activity, a circle with a given radius for a *Circular* activity, or an irregular shape depending on the route.

To determine the spatial and temporal contexts of commuting activities, we use the Open Source Routing Machine [OSRM, [59]]. We implemented a separate Python class *Routing*, that is instantiated with separate datasets which correspond to different commute modes. These datasets can be generated with OSRM tools from OpenStreetMap [60] datasets. The modeling framework supports commute modes by car, foot, bicycle, or train. To obtain the spatial context of a commute route, we implemented a

```

1 class CommuterWeekday(Diurnal):
2     def construct(self):
3         # Initialise OSRM and collection of all diaries
4         self.r = Routing("car.osrm", "bike.osrm",
5                          "foot.osrm", "public.osrm")
6         diaries = Diaries(self.t_start, self.t_end)
7         for agent_id in agents:
8             h_dur = duration("7:30")
9             w_dur = uniform("6:30", "7:30")
10            dist, c_dur = self.r.distance(x1, y1, x2, y2, "car")
11            act_1 = Location("home", h_dur, x1, y1)
12            act_2 = Route("commute", c_dur, x1, y1, x2,
13                          y2, "car")
14            act_3 = Circular("work", w_dur, x2, y2, 500)
15            diary = Diary(agent_id)
16            diary.add(act_1)
17            diary.add(act_2)
18            diary.add(act_3)
19
20 def calculate(realisation): # Fixed random seed
21     c = CommuterWeekday(default_rng(seed=13), realisation)
22     c.time("2020-07-01 00:00:00", "2020-07-02 00:00:00")
23     c.generate()
24     c.enrich("utrecht.lue", [["NO2", "PM25"], ["noise"]])
25
26 for idx in range(20): # Run 20 Realisations
27     calculate(idx)

```

Fig. 2 Python code excerpt demonstrating the specification of activity duration, different activity types, and the composition of a diary. x1, y1, and x2, y2 indicate coordinates for residential and work locations. The assigned commute mode is by car. The remainder specifies a diurnal cycle for a commuter in the CommuterWeekday class, initializes a simulation in the calculate function, and executes a set of 20 realizations.

method that takes coordinate pairs for origin and destination, and a commute mode. It queries the OSRM and returns a rasterized route matching the spatial discretisation of the exposure attributes. Exposure values for commuting can then be extracted by basic overlay. For the temporal context we added a method that provides the distance and estimated duration of a route for a particular commute mode. This function can be used to determine the durations of commute activities while generating an agent schedule.

The *exposure* framework class combines the information stored in the activity diaries and spatial context classes to calculate the final exposure values. First, all time steps to assess the exposure are determined. Exposures need to be reassessed when either the activity of an agent changes or the attribute values of an environmental factor change. For each agent, the modeling framework intersects a diary with its particular time stepping and the time stepping of the environmental attributes. If required, an activity is subdivided and added to the activity diary. Afterwards, for each activity, the spatial context is determined, overlaid with the environmental data, aggregated over the affected area (i.e., buffer or route), and assigned as exposure value for the corresponding activity. In a final step, the time-weighted exposures per activity and diurnal cycle are calculated.

CASE STUDY

We demonstrate the concepts and software framework described in the previous sections by means of a proof of concept study for the population in the province of Utrecht (8% of the total Dutch population) in the Netherlands, covering rural and urban areas. We illustrate how to calculate exposure estimates to air pollution (NO_2 , $\text{PM}_{2.5}$) and noise. Additionally, we describe the development of two personal exposure data sets for the MOBI-AIR study for Switzerland and the Netherlands.

Study area and data sets

The province of Utrecht is located in the center of the Netherlands (Fig. 3). The province covers an area of about 1500 km² with about 1.4 million inhabitants, corresponding to ~4% and 8% of the Netherlands' total area and total population, respectively. About 360,000 inhabitants live in the city of Utrecht.

We used the national cadastral dataset [61] containing the front door coordinate locations and the functional classes of intended use of all buildings in the Netherlands. We used locations with "residential function" as residential address locations, thereby covering residential and partly residential buildings, and extracted the locations for the province Utrecht. We selected buildings classified as commercial, healthcare, office, education, industry, hotel, or sport as potential working locations. We excluded temporary buildings or auxiliary constructions in our analysis. To allow for work locations outside the province, we included all work addresses in the Netherlands. Overall, we considered 626,381 residential addresses located in the province as our study population, and nationwide, 1,057,999 potential work locations.

The traffic infrastructure, consisting of road networks, railway tracks, cycling and walking paths, was derived from OpenStreetMap [60] data, obtained as January 2022 country snapshot [62]. We used the Open Source Routing Machine [OSRM, [59]] to extract different routing networks to allow for commuting either by car, bike, train, or walking.

We calculated exposure to three different environmental factors, noise and the air pollutants NO_2 and $\text{PM}_{2.5}$ (Fig. 3B–D). For air pollution, we used annually averaged maps with an hourly time step, i.e., 0–23 h [56]. The datasets provide concentrations in $\mu\text{g m}^{-3}$ for weekdays and weekend days (Fig. S1–S4). For exposure to noise, we used the average noise estimates by the Dutch National Institute for Public Health and Environment, providing the annual average day-evening-night noise level, with a respective 5 and 10 dB penalty for the latter two periods (Lden, in dB; Fig. 3B; RIVM, 2022). This single data set provides combined noise levels emitted by road traffic, air traffic, railway traffic, industrial areas, and wind turbines for the average 24 h day and therefore does not allow for distinguishing between different hours, or weekdays and weekend days. The air pollution surfaces were used at the original resolution of 25 m, and the noise raster was resampled from 10 m to 25 m to reduce the computing time. All datasets provide nationwide coverage.

Activity profiles for homemaker and commuter

We developed a set of characteristic diurnal activity diaries as behavioral input for the modeling framework (see Table S1). These diaries can be applied for workdays (wd) and weekend (we) days. The H_{wd} and H_{we} diaries include an outside activity around the residential address to model, e.g., leisure or shopping activities, the remainder of the day, the agent stays at the residential location. We assigned a two-hour outside activity sometime between 8 a.m. and 10 p.m. within a buffer of a 10 km radius. The C_{wd} diary includes 8 h spent at work, 2 h for commuting back and forth to work, and staying the remaining time at the residential location. We assume that commuting only occurs during weekdays.

As we do not have detailed knowledge about the agent's work addresses or their commute modes, we assign work locations by using destination areas derived from survey data. For C_{wd} we use national travel survey data for the years 2011–2017 from Statistics Netherlands [63], including 181,489 commute trips, to derive an origin-destination matrix determining potential work areas based on postal code areas. We used the first two digits of the postal code to define origin and destination areas (see Fig. 3A) as these contain a sufficiently large number of travel survey participants. A more detailed description of the travel survey input data and classifications used for the origin-destination matrices is given in the Supplementary Material Section 2. Within a chosen destination area, we randomly select a work address. Then, the Euclidean



Fig. 3 Maps showing the province of Utrecht study area. **A** the residential addresses (green) located in the province of Utrecht, the potential work addresses (orange), the boundaries between the postal code areas (gray), and the municipality boundary for the city of Utrecht (red). The inset map shows the location of the province, **B** yearly averaged noise levels (RIVM, 2022), **C** NO_2 concentration at 8 a.m. weekdays [56], **D** $\text{PM}_{2.5}$ concentration for 8 a.m. weekdays [56].

distance between residential address and work address is calculated and used to look up the probabilities for each commute mode for that particular distance range. The distance ranges can be used to specify that, for instance, the walking and cycling modes are more likely selected than the car mode when the distance to a work location is <1 km. Then, one of the four commute modes is determined based on the probabilities given. For the selected commute mode, the route along the traffic network is determined. If the estimated commute time exceeds the maximum duration of the commute activity, the procedure is repeated with a new destination location; otherwise, the track is overlaid with the exposure maps to calculate the exposure values while commuting.

Finally, we do not know the type of person living per residential location; therefore, we calculate all profiles for all locations. We ran 75 realizations each for the H_{wd} , H_{we} , and C_{wd} diaries, and calculated according to Eq. 2 the exposure estimates for a *homemaker* as $H = 5 \cdot H_{\text{wd}} + 2 \cdot H_{\text{we}}$ and for a *commuter* as $C = 5 \cdot C_{\text{wd}} + 2 \cdot H_{\text{we}}$. Simulation outputs are ensemble mean and standard deviation per exposure for each residential location and each profile. These values can be used for further analysis.

RESULTS

Figure 4 shows for all realizations the resulting durations of the activity diaries and the corresponding diurnal exposure estimates for NO_2 . The majority of travel times for one commute trip are <30 min, as most of the trips are assigned within the residential postal code area.

Figure S7 shows the probabilities of selected destination areas for a single *commuter* agent, according to the origin-destination

matrix for agents residing in the city of Utrecht. The majority of resulting commute routes remain in the city of Utrecht.

The simulation output can also be extracted to investigate ensemble statistics of the entire study population, as shown in Table S2 and Fig. 5, where we evaluate different activity patterns and their influence on the exposure estimates. For the comparison of the *homemaker* and *commuter* profiles, for example, a homemaker with low ($<20 \mu\text{g m}^{-3}$) estimated NO_2 exposure is likely to be underestimated compared to a working person. For high ($>40 \mu\text{g m}^{-3}$) values, homemaker exposures are likely overestimated. An explanation can be that people living in less polluted urban locations work in more polluted areas. On the contrary, people living in the more polluted areas tend to work in less polluted locations and consequently results in lower exposure estimates.

The modeling framework propagates uncertainties in the model inputs to outputs, i.e., exposure estimates also contain uncertainties [52], Chapter 2]. For *commuter* agents, the uncertainty is ~ 1 –2 standard deviation ($\mu\text{g m}^{-3}$), as shown in Fig. 5D. This uncertainty is mainly caused by uncertainty of the exposure during work and the commute trip, both due to the uncertainty in the work location within the destination regions of commuters.

Location information of residential addresses and exposure estimates can also be combined to investigate the spatial pattern of the predictions. Figure 6, for example, shows the difference calculated between *homemaker* and *commuter* agents for residential locations in the city center of Utrecht. In general, residents living along roads with high traffic intensity show reduced exposure estimates, while low pollution residential locations show increased exposure estimates when commuting is considered. Figures S8 and S9 show the corresponding differences for $\text{PM}_{2.5}$ and noise, respectively.

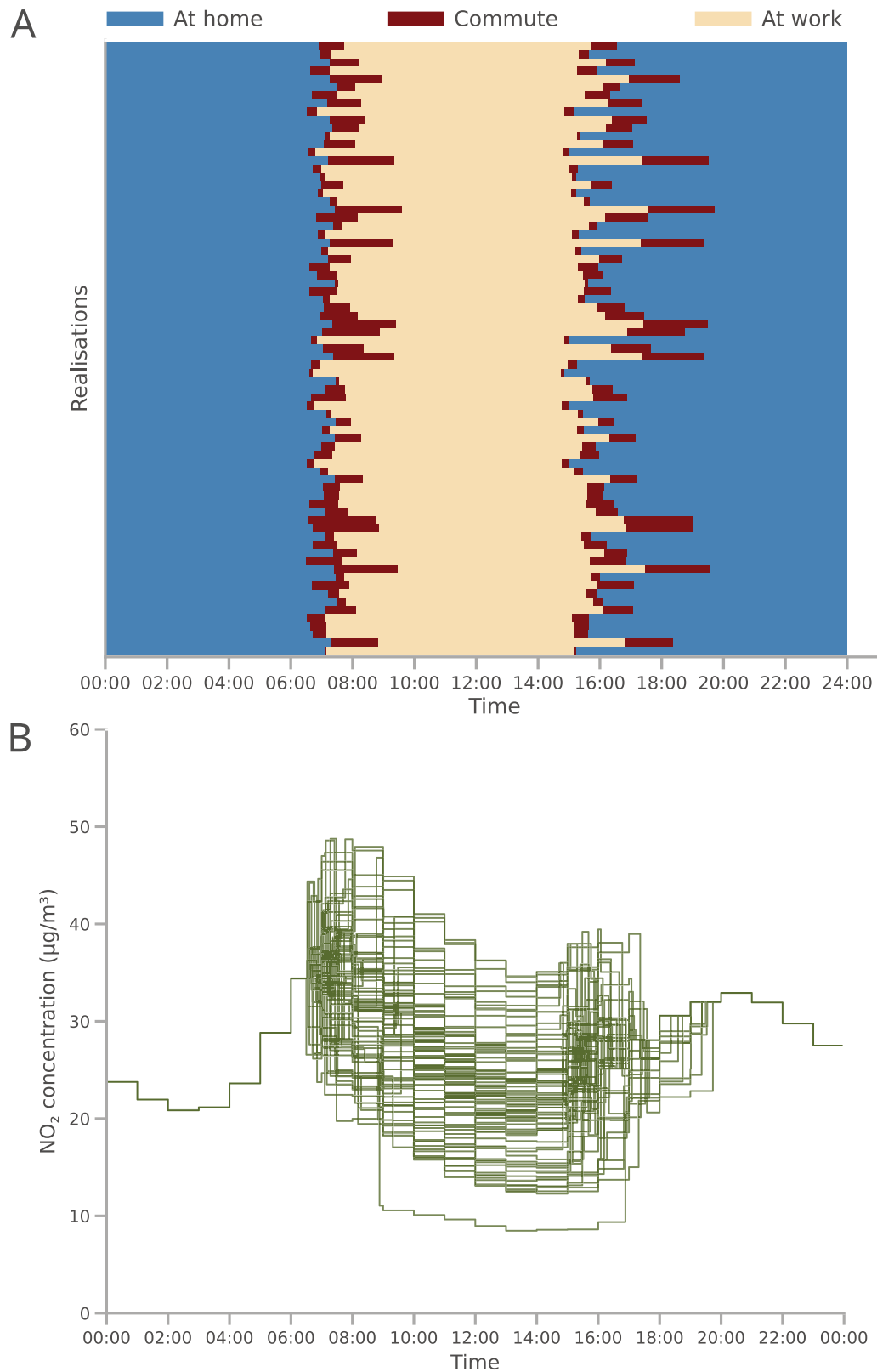


Fig. 4 Results shown for 75 realizations, selected for a single commuter agent. **A** Durations of the activities (in minutes). **B** Variation of the diurnal exposure estimates for NO₂.

NO₂ and PM_{2.5} exposure estimates for Switzerland and the Netherlands

As part of the MOBI-AIR project, we applied the modeling framework to create a data set of personal exposures to NO₂ and

PM_{2.5} for all residential addresses in Switzerland, which were linked to the participating cohorts. From the Federal Register of Buildings and Dwellings [64], we derived 1,867,667 residential and 1,334,473 potential working locations. The Swiss Structural Survey

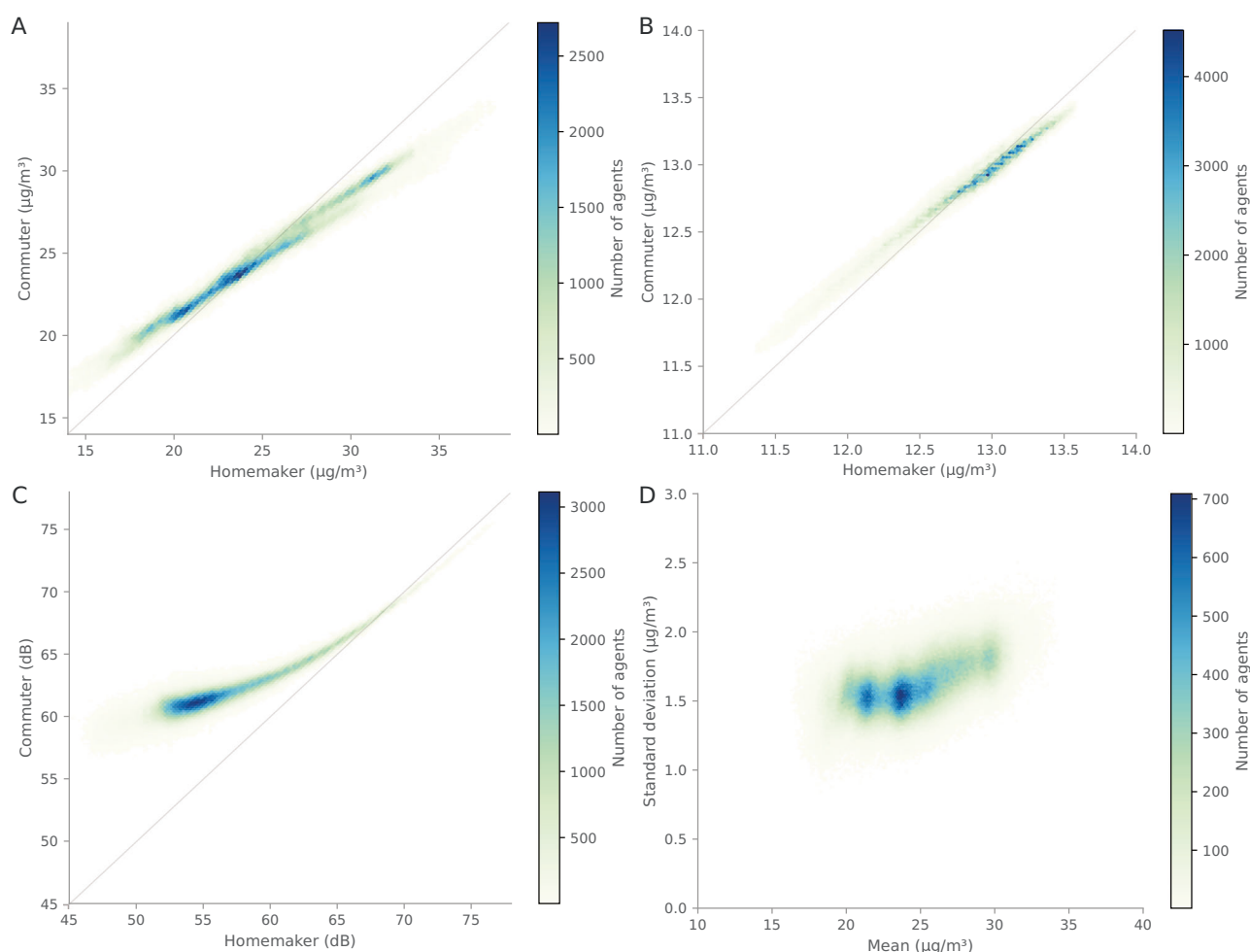


Fig. 5 Comparison of homemaker and commuter for 626,381 address locations in the province of Utrecht. A NO₂ values. **B** PM_{2.5} values. **C** Noise values. **D** Uncertainty in personal exposure values shown as standard deviation of personal exposure plotted against estimated personal exposure, shown for the NO₂ commuter exposure estimates.

[65] was used to derive origin-destination matrices that were further stratified by sex and into low, intermediate, and high socioeconomic status. The areas for origins and destinations correspond to the administrative level of municipalities. A more detailed description of the travel survey and classifications used for the origin-destination matrices is given in the Supplementary Material Section 2.

Models were developed to convert nationwide annual NO₂ and PM_{2.5} surfaces [66–68] at 100 m resolution into long-term hourly NO₂ and PM_{2.5} concentration surfaces. The models were based on analysis of daily NO₂ and PM_{2.5} monitoring data across Switzerland at 63 and 9 monitoring stations, respectively, representing urban, suburban, and rural background locations. The obtained concentration surfaces were then used to estimate weekday and weekend day exposures.

As the type of person is unknown for each residential location, we ran 50 realizations for all locations and each activity diary to compile the exposure estimates. Figure S10 shows the comparison of mean NO₂ values for the homemakers and commuters. Comparable to the results shown for the Utrecht case study, a homemaker with <10 µg m⁻³ estimated exposure is likely to be underestimated compared to a working person. For values above 40 µg m⁻³, homemaker exposures are likely overestimated. The uncertainty is ~1–2 standard deviation (µg m⁻³) for *commuter* with work locations sampled from origin-destination matrices (Fig. S10C). Additionally, we calculated exposure estimates for

participants of the SAPALDIA study [47–49] where known work address locations were available.

For the Netherlands, we used the same NO₂ and PM_{2.5} data as described in the case study. We also stratified the travel survey data by sex and into low, intermediate, and high socioeconomic status to derive the origin-destination matrices. See the Supplementary Material Section 2 for a more detailed description. For all residential locations of the EPIC-NL cohort [50] and each activity diary, we ran 50 realizations to compile exposure estimates.

DISCUSSION

We introduced the concepts and a prototype implementation of a framework for personal exposure assessment for multiple environmental factors. We demonstrated the capabilities in a case study for the population of the province of Utrecht, where we evaluated different activity profiles in probabilistic mode against three different environmental exposures. The modeling framework was also used to create data sets of personal exposures for a large number of residential addresses in Switzerland and the Netherlands, with the emphasis to study the associations between exposures and health effects [e.g., [51]].

With the case study, we demonstrate the capabilities of the framework and illustrate further processing of the estimated exposures. No measured personal exposures were available for our study population; the exposure estimates, therefore, lack the

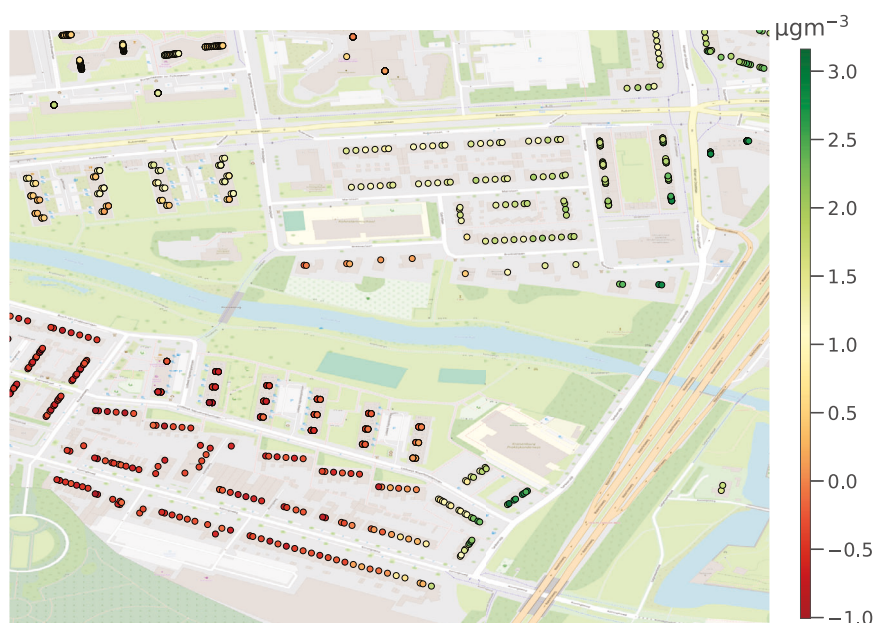


Fig. 6 Differences in NO_2 exposure estimates between homemakers and commuters are shown for residential addresses in the city of Utrecht. OpenStreetMap [60] is used as background.

same degree of accuracy as obtained in short-term experimental studies performed with high-precision personal exposure monitoring [69]. The exposures estimated by the modeling framework likely include a higher uncertainty compared to directly measured exposures, but our framework also allows to quantify the uncertainties in the exposure estimates.

The air pollution exposures estimated by the modeling framework presented in our manuscript using the commuter profile were compared to estimated exposures using the same air pollution surfaces and recorded daily time activity diaries and routes obtained by GPS trackers [[70], in review]. For 489 participants in Switzerland and 189 participants in the Netherlands, high correlations for NO_2 ($R^2 = 0.81$ for the Netherlands and $R^2 = 0.77$ for Switzerland) and moderate to high correlations for $\text{PM}_{2.5}$ ($R^2 = 0.52$ for the Netherlands and $R^2 = 0.80$ for Switzerland) were found.

The environmental input datasets used in the case study are based on previously published studies in which validation has been done. For the development of the Dutch air pollution surfaces, air quality monitoring data from 544 stations for NO_2 and 227 stations for $\text{PM}_{2.5}$ in the Netherlands and the neighboring countries Belgium and Germany were used. A detailed development and evaluation of the Dutch air pollution surfaces is described in the publication of [56]. For Switzerland, 63 background stations for NO_2 and 9 background stations for $\text{PM}_{2.5}$ were used. The Swiss air pollution surfaces are described and evaluated in the publications of [67, 68]. The noise map was created by the Dutch National Institute for Public Health and the Environment [71] using the Standard Model Instrumentation for Noise Assessments model. The residential and work locations are derived from Dutch cadastral data [61]. Several software tools used by the framework ([e.g., [58, 59, 72, 73]]) are well tested with respect to their functioning. Own software components developed during the project were tested individually.

The modeling framework provides building blocks to define activity profiles for cohort participants and to subsequently assess personal exposures to one or more environmental factors. We defined activity diaries with typical diurnal behavior, and used those diaries to represent commuters and homemakers. We used travel survey data to derive origin-destination matrices and, with

that, refined the arbitrary allocation of potential work locations. Compared to homemaker exposure estimates, the location and environmental factor concentrations of workplaces affect the estimates and can result in a high spatial variation for agents living within a densely populated area, as illustrated in Fig. 6.

In general, the modeling framework can be applied to a different study area by using freely available datasets. OpenStreetMap data provides global coverage and can be used to derive several required input datasets for a particular study area. Routing networks for different commute modes can be extracted from OpenStreetMap, as shown in the case study. Locations of residential and work locations can be derived from the OpenStreetMap building layer instead of using cadastral datasets if these are not available. However, modelers need to be aware that for a large proportion of buildings on OpenStreetMap, the building type is unknown [74] and would need to be classified. If this proportion is too large for a particular area, other data sets, most likely regional [e.g., [75]] or local data, will be needed to better determine a building type. Also, the quality of OpenStreetMap data should be sufficient for their study area [e.g., [76–78]].

This framework can be used for assessing additional exposures, such as temperature or green space, as long as the datasets have a sufficient spatial and temporal resolution. The datasets need to contain values that are representative for the estimation of long-term exposure.

In our case study, we assessed the personal exposure of humans to multiple environmental factors. The modeling framework could also be used for assessing animals by assigning activity duration, e.g., spent in a shed and foraging locations. The resulting exposures provide information on animal health and could be used to improve planning, intervention, or policy-making, preventing contamination of, e.g., bees [e.g., [79]], grazing livestock [e.g., [80]], or marine organisms [e.g., [81]].

Activity profiles for agents could follow the shown approach from general to more specific profiles, and can be directly implemented if the available spatial contexts are sufficient, for example, the type of work was known for blue vs. white collar workers. Separate activity profiles for toddlers, pupils, or other population groups can be developed comparatively. In general,

the development of new commuting profiles involves the following steps: first, the destination locations specific for a target group need to be identified, such as locations of kindergartens, schools, office buildings, or factories. Then, distance ranges and probabilities for the different commuting modes to the destinations need to be refined. Finally, the duration at of the stay at the destination location and the starting times of those activities need to be assigned. The exposure assessment can then be performed with the activity profile. We used origin-destination matrices to constrain the assignment of work locations for our study population. However, a fully randomly drawn work location could be used if no origin-destination matrix is available to refine the work assignment, or daytime vs nighttime population data could be used to mimic mobility patterns [e.g., [82]]. In our examples, we used up to five activities in our profile descriptions. More activities could be included if information about residence and duration is available, although additional activities will increase the overall simulation time. Additional data included in travel surveys could be further used to improve parameterizing the start times of a commute activity or to more precisely assign commute modes per age group.

In our study, we accounted for uncertainty in exposure estimates in time by using probability distributions for start and end times or durations of agents' activities, and in space by drawing possible work locations from origin-destination matrices. We did not account for uncertainty in the environmental factors. This can be achieved by using several realizations of, for example, air pollution data instead of assessing a single dataset. In general, other required input arguments could be randomly parameterized as well, such as the radius for a buffer activity. We used diurnal air pollution surfaces for weekdays and weekends. We did not use the seasonality already expressed in the NO₂ and PM_{2.5} datasets [56], as no behavioral data was available to parameterize separate profiles for the warm and cold seasons and to reduce the input data size required for the case study. The estimated exposures, therefore, neglect seasonal or monthly variation in the air pollution surfaces and human activity patterns. If seasonal or monthly input data is available, input for environmental factors and a set of profiles can be created for each time period, e.g., the month of the year. Afterwards, the exposure assessment framework can be executed twelve times to capture the monthly variation in the exposure estimates. A disadvantage of this approach is a considerable increase in compute and storage demands.

We used hourly air pollution data as an example for dynamic environmental factors, and one input for noise as an example for a static factor. Depending on the type of environmental factor, a higher temporal resolution may be required, such as for noise, where dynamic data is more appropriate to capture the day, evening, and night noise levels. The spatial discretization used for the environmental factors, in our case study 25 m, will also affect the exposure estimates. Using higher resolution for the environmental factors could better capture a distance-based variation of pollutants emitted, e.g., from road sources, and therefore improve the exposure estimates. Disadvantageous is again an increased demand for storage and compute.

For commuting, we estimated exposures using the shortest routes, therefore neglecting that agents could deviate, for example, due to personal choices [e.g., [83, 84]]. If available, additional waypoints could be added to the routing engine to refine the returned path. The estimated duration of commute activities for walking and cycling is based on the default speeds set per commute mode by OSRM. Those speeds might not be representative when modeling, for instance, children or elderly persons. The default speeds, however, can be adapted when creating the input files for OSRM. In general, the duration of a commute activity is estimated based on the uniform speed of a commute mode and the network distance returned by OSRM.

Additional time caused, for instance, by lower actual speeds due to high traffic densities or by additional stops at traffic lights is therefore neglected and may lead to underestimated commute activity durations. An additional factor could be added to the modeling framework to incorporate an adjustment of the individual commute mode durations. OSRM supports the main commute modes by default, but does not provide routing specifically for public transport. An important commute mode is therefore not represented in the commuter profiles and consequently in the exposure estimates.

We also did not incorporate indoor-outdoor ratios when calculating the exposure estimates per activity, as individual information on building penetration rates determining the indoor-outdoor contrasts was not available. However, studies show that including indoor-outdoor pollutant ratios can improve exposure estimates [e.g., [85]]. If these ratios are known or can be estimated, they can be used as factors in the time-weighted calculations of the activity exposure estimates. We also assume that the whereabouts of agents are equally distributed in space by calculating the average exposure values in the spatial contexts for circular areas and routes. With limited effort, weighted buffer calculations such as Gaussian decay functions could be added to the framework to better represent distance decay relationships with the environment [54, 86].

We illustrated that exposure estimation for hundreds of thousands of people can be done with reasonable computational effort. For Switzerland, we used the framework to estimate NO₂ and PM_{2.5} exposures for about 1.87 million residential addresses and 13 profiles, with runtimes for a realization of a commuter diary (C_{wd}) of about 31 h on a single CPU of an AMD EPYC 9654. As realizations are independent of each other, multiple runs can be executed in parallel to reduce the overall runtime for Monte Carlo simulations. Another opportunity to reduce the runtime is to preprocess activities having large buffer sizes by storing the aggregated values in a raster and then lookup cell values at runtime instead.

Instead of using web services for querying routes, we developed a Python module for the OSRM routing engine and used OpenStreetMap snapshots as input data. With that, it is possible to generate reproducible results for the commute routes. In addition, an unlimited number of routing requests, independent of web access and usage limitations for querying a service, can be executed locally. Additional requests would also be required if the commute mode would be assigned as a result of network instead of Euclidean distance.

CONCLUSIONS

We presented a software framework for the development and assessment of simulation models for long-term large-scale exposure assessment. We considered different activity profiles representing typical behaviors of individuals, thereby including mobility in their daily activities. We showed the application of the framework for uncertain scenarios, where results indicate that incorporating commuting and work locations can differ significantly from residential estimates for individuals. The modeling framework provides a consistent way of calculating exposures to multiple environmental factors. With minor modifications and by using publicly available datasets, the modeling framework can be used to model other study areas or environmental attributes.

SOFTWARE AVAILABILITY

We published the scripts to set up a Python environment and the model scripts demonstrating the executable workflow of the presented case study at https://github.com/computationalgeography/paper_agent_based_exposure_assessment. The modeling framework is free and open-source software under the MIT

License and developed at https://github.com/computationalgeo/graphy/agent_based_exposure_assessment.

DATA AVAILABILITY

Data will be made available on request.

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AUTHOR CONTRIBUTIONS

Oliver Schmitz: Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing—original draft, Writing—review and editing, Visualization. Kees de Hoogh: Conceptualization, Methodology, Writing—review and editing, Project administration, Funding acquisition. Nicole Probst-Hensch: Resources, Project

administration, Writing—review, and editing. Ayoung Jeong: Data curation, Writing—review, and editing. Benjamin Flückiger: Resources, Data curation, Writing—review, and editing. Meng Lu: Methodology, Writing—review, and editing. Aisha Ndiaye: Data curation, Writing—review, and editing. Danielle Vienneau: Resources, Data curation, Writing—review, and editing. Gerard Hoek: Conceptualization, Methodology, Writing—review and editing, Project administration, Funding acquisition. Kalliopi Kyriakou: Writing—review, and editing. Roel C. H. Vermeulen: Writing—review and editing, Project administration, Funding acquisition. Derek Karssenbergh: Conceptualization, Methodology, Formal analysis, Investigation, Writing—original draft, Writing—review and editing, Supervision, Project administration, Funding acquisition.

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COMPETING INTERESTS

The authors declare no competing interests.

ETHICS APPROVAL

This research does not involve human subjects or animal research.

ADDITIONAL INFORMATION

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Correspondence and requests for materials should be addressed to Oliver Schmitz.

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